

EDA U-Net: Effective Dual Attention U-Net for Medical Image Segmentation

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Abstract—The contemporary challenge in computer vision and image processing lies in achieving precise and reliable binary segmentation in medical images such as eye and skin lesion datasets. Many existing studies suffer from inaccuracies in segmentation, resulting in insufficient feature extraction and inappropriate classification, which is particularly critical in the medical field. This paper proposes a methodology that significantly advances binary image segmentation tasks by seamlessly integrating dual attention mechanisms into the well-established U-Net architecture. Our approach strategically positions the dual attention module at the critical junction between the encoder and decoder sections, utilizing both position attention and channel attention modules to enhance feature representation. The validation process through comprehensive experiments on benchmark datasets demonstrates performance improvements over conventional U-Net architectures and six other state-of-the-art deep learning models. The proposed model achieved a mean Intersection over Union (mIoU) of 96.13% on the eye disease dataset and 96.17% on the skin lesion dataset, consistently outperforming the other models. The model's performance, when compared to multiple other methods, indicates it is an effective solution. The automated segmentation process could help reduce the time and effort required by medical professionals, contributing to more efficient use of their expertise across various fields

Index Terms—Binary Image Segmentation, Dual Attention Mechanism, U-Net Architecture, Medical Image Analysis, Deep Learning

I. INTRODUCTION

COMPUTER vision is a vital field within artificial intelligence that focuses on enabling computers to understand and interpret visual data from their surroundings. Its importance has been increasing in recent years due to the growing volume of visual information produced by digital devices. Applications of computer vision span various areas, including facial recognition [1],[2], object detection [3]-[5], autonomous vehicles [6],[9], recommendation systems

[7],[8], and, notably, diagnostic classification in the medical field. Numerous techniques have been developed to extract meaningful information from visual data, which can be used for decision-making or classification tasks. Deep Learning (DL) algorithms play an important role in these techniques, assisting computers in recognizing patterns and features within images or videos to achieve these goals. Recently, key technologies in computer vision, such as image segmentation [10]-[12], image classification [13]-[16], and object recognition [17]-[19],[22], have been widely employed. As the field of computer vision has advanced, the development of computer-aided diagnostic systems offering valuable support to healthcare professionals in identifying and diagnosing various medical conditions are attainable with current technologies for analyzing visual data [20],[21]. Especially, DL techniques have been especially intensely utilized to advance computer-aided diagnostic systems for disease diagnosis, contributing to heightened efficiency and accuracy in the diagnostic process [23].

Despite advancements in computer vision, machine learning (ML), and deep learning (DL), challenges remain in achieving high accuracy and reliability when diagnosing visual data, such as images and videos. Among the diverse processes, segmentation is important for focusing on specific regions of interest (RoI) or isolating objects within images for more precise analysis and diagnosis [24]. It affects the performance in dealing with diverse objects, refining boundaries, and ensuring the robustness of segmentation models. Binary segmentation techniques have been introduced to improve computer-aided medical diagnosis by identifying relevant areas within medical images [25]. These deep learning strategies have proven useful in diagnosing and classifying various diseases, as they can extract complex patterns and features to identify RoI, object boundaries, or specific regions within medical images. This information is then used for classification tasks to aid in diagnosis. As a result, these approaches have contributed to improving the precision and efficiency of diagnostic processes.

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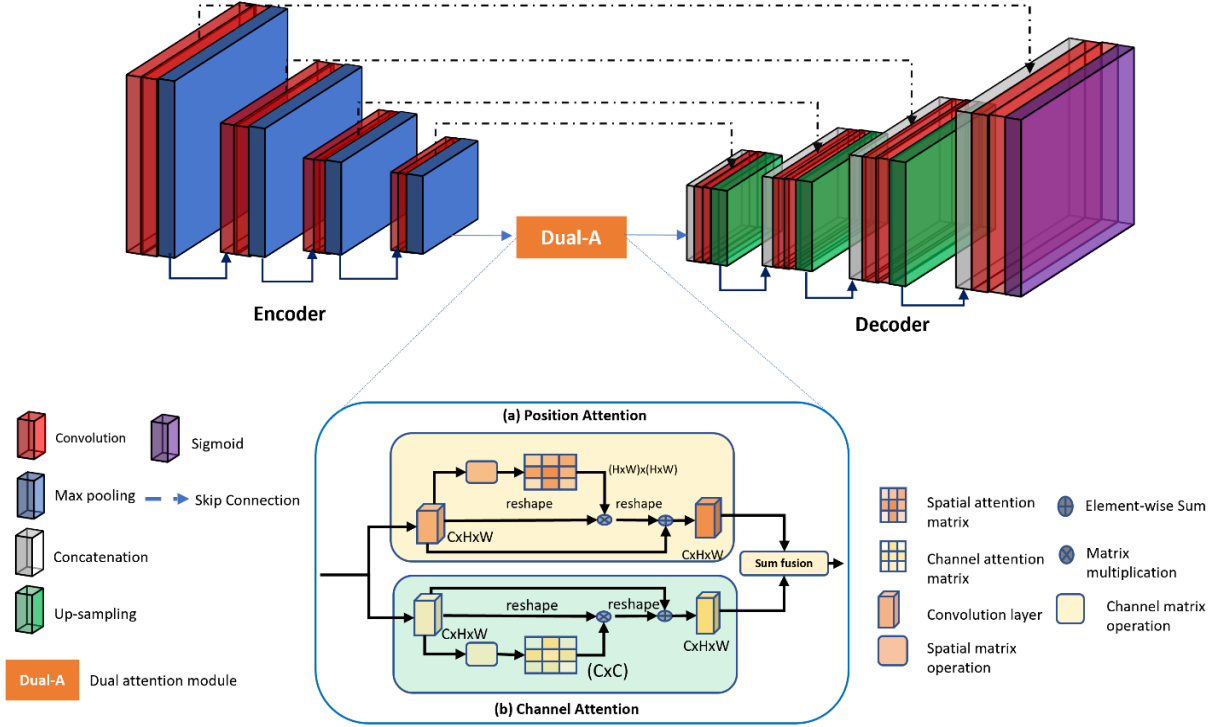


Fig. 1. The proposed model architecture of the integration of U-Net with a dual attention module, incorporating both (a) position attention and (b) channel attention.

However, challenges still exist in achieving high precision and robustness in segmenting objects or regions within images, particularly when dealing with diverse datasets and refining segmentation boundaries. Limitations such as the availability of sufficient datasets, lower segmentation accuracy, and issues with feature extraction and classification accuracy persist. Therefore, there is a need for improved binary segmentation methods that can address these challenges and deliver better performance in terms of accuracy and efficiency to meet the growing demands of computer vision tasks.

In this paper, we propose a medical image segmentation method based on an enhanced dual attention U-net architecture for improving the segmentation performance of eye disease and skin lesion datasets. The main objective of this research is to enhance the practical approach for accurate and reliable binary segmentation reducing complexity compared to existing studies in eye and skin lesion datasets. To achieve this objective, the proposed method integrates the U-Net, with a dual attention technique comprising position attention and channel attention. In this design, the final convolutional layer of the encoder functions as the input to the Dual Attention module, with its output seamlessly transitioning into the initial convolutional layer of the decoder block. This integration optimizes the focus on relevant feature extraction within the image, potentially enhancing feature extraction capability and leading to more accurate and robust binary segmenting results with less complexity. The performance of the proposed method is evaluated based on the datasets, such as eye disease datasets and skin lesion datasets.

Additionally, we performed a comparative assessment against state-of-the-art methods for these two datasets. The rest of the paper is organized into 7 sections. Section 2 presents the related work. In Section 3, we describe the proposed method, while Section 4 provides details about the datasets and implementation. Section 5 illustrates the results obtained using the proposed techniques and includes a comparison with other existing methods. Section 6 offers a discussion of the findings. Finally, Section 7 presents the conclusion and future work.

II. RELATED WORK

A. Standalone Deep Learning Architectures

In computer vision, the segmentation technique is crucial, with diverse applications in various domains such as medical imaging, autonomous driving [9], and object recognition [24]. Both traditional and advanced methods have been proposed in the field of deep learning-based approaches for segmentation tasks. These developments aim to enhance the accuracy and efficiency of segmentation processes, marking a critical evolution in the application of deep learning techniques to image analysis.

Traditional binary segmentation methods, including thresholding techniques [26], edge detection techniques [27], and region-based techniques [28], have served as the foundation for image segmentation. While their simplicity and computational efficiency make them accessible for basic segmentation tasks, these methods exhibit notable limitations. These constraints may include challenges in handling complex

structures, variations in image quality, or difficulties in capturing fine details. These methods often require parameter fine-tuning to optimize their performance, which may struggle to accurately delineate objects with complex shapes and structures in the image. Fixed threshold values can limit their adaptability to diverse image contexts.

In recent years, advanced deep learning-based architecture, such as Fully Convolutional Networks (FCNs) [29], DeepLab [30], U-Net [31], and SegNet [32], have emerged as robust solutions in the field of image segmentation. These architectures enable the learning and adaptation to various characteristics within images, demonstrating significant advancements in image analysis and segmentation tasks. These methods indicate exceptional capability in addressing intricate segmentation challenges, proving indispensable in object detection, medical image analysis, and scene parsing. The transition towards advanced techniques reflects the dynamic evolution of image segmentation, driven by the investigation for heightened precision and adaptability in handling diverse visual data. Especially one of the breakthroughs in image segmentation is the U-Net architecture. The remarkable achievements of U-Net can be attributed to its unique capability to capture both local and global features effectively. It is achieved through its innovative design, incorporating a contracting path as an encoder and an expansive path as a decoder [29]. It helps to learn detailed information while maintaining a broader contextual understanding, making it well-suited for image segmentation tasks. However, this architecture also exhibits an issue in fine-grained localization, as up-convolutions in the decoder may lead to the loss of precise spatial information. Additionally, the emphasis on local features limits its capacity to take broader contextual information, especially at lower resolutions, affecting the model's ability to capture long-range dependencies in the data.

B. Deep Learning Integration with Attention

Recently, various studies have proposed incorporating advanced methods with additional modules to enhance the model's capabilities. This approach is often employed to address specific challenges with the aim of improving the model's overall performance and adaptability. Attention mechanisms, in particular, have raised considerable attention for their ability to enhance the performance of deep learning models [33]. These mechanisms enable models to dynamically assign varying levels of importance to different regions and channels within an input.

A seminal work introduced the Transformer model [34], effectively employing self-attention mechanisms for processing sequential data, such as natural language. Notably, attention mechanisms have found valuable applications in computer vision tasks, where they enhance the model's ability to focus on relevant features, contributing to improved performance in tasks such as image classification and object detection. The integration of attention mechanisms into image segmentation models has shown promise in improving the accuracy of object representation or object detection.

Attention U-Net was proposed, incorporating spatial

attention gates to refine segmentation tasks [35]. This innovative approach allows the model to emphasize important spatial regions selectively, contributing to more accurate and refined segmentation outcomes. Similarly, the Convolutional Block Attention Module (CBAM) was introduced as a significantly advanced model. CBAM dynamically recalibrates feature maps, forwarding improved object localization [36]. This dynamic recalibration mechanism enhances the model's ability to focus on relevant features, contributing to heightened precision in object localization tasks.

In addition, a recent advanced method in image segmentation is the Dual Attention Network (DA-Net), which is proposed to enhance segmentation tasks with improved attention mechanisms [37]. DA-Net consists of two attention modules to refine the extraction feature from the network backbone. Position Attention, proposed as a mechanism adept at selectively attending to spatial relationships within images, and Channel Attention, emphasizing the significance of feature channels, have become notable components. Combining these dual mechanisms enables the capture of fine-grained spatial and channel-wise information, enhancing feature representation capabilities.

C. Medical Image Segmentation

In the domain of medical image segmentation, a necessary and dynamic landscape of methodologies has emerged in recent years. Traditional approaches, which include thresholding, region growing, and region-based techniques, have historically laid the foundation for understanding structures and specific regions with image to analysis. The advent of ML marked a significant transition, with early techniques like support vector machines (SVM) and decision trees exploring the automation of segmentation [38]. However, these methods faced limitations in handling the complexity and variability of medical images, often resulting in unacceptable accuracy.

The field underwent a transformative shift with the introduction of deep learning. Approaches such as Convolutional Neural Networks (CNN) and specialized architectures like U-Net, Res-Unet [39], DeepLab, and Attention U-Net revolutionized medical image segmentation. These advanced deep learning methodologies have been extensively employed across diverse medical datasets, showcasing their efficacy in various conditions. Additionally, the segmentation process has gained depth by applying advanced techniques, including the fusion of multimodal images and the utilization of transfer learning from pre-trained models [40],[41]. Binary segmentation, vital for delineating anatomical structures and detecting pathological regions or regions of interest (ROI) in disease areas, has demonstrated utility in critical domains like tumor detection, organ segmentation, and brain image analysis [42]. Medical image segmentation competitions and benchmark datasets have documented challenges and showcased advancements, providing a platform for evaluating novel methods. Standard evaluation metrics such as the Dice coefficient and sensitivity serve as benchmarks for assessing segmentation accuracy

[43]. State-of-the-art methods have exhibited innovations, contributing to enhanced precision and reliability of segmentation in clinical contexts. These advancements have practical implications, impacting healthcare by aiding disease diagnosis, treatment planning, and patient monitoring.

III. PROPOSED METHOD

We present the Dual Attention U-Net architecture, developed to improve the precision of binary image segmentation, especially in the domain of medical images. The architecture is based on the well-known U-Net structure, enhanced by the integration of dual attention mechanisms, specifically designed to capture both spatial and channel-wise relationships in feature maps. These attention modules are inserted at the key junction between the encoder and decoder sections of the network, as shown in Figure 1. This design enables our model to learn and utilize both spatial and channel information effectively, leading to enhanced feature representation and, consequently, superior segmentation performance.

A. U-Net as the Foundational Framework

The core of our model is built upon the U-Net architecture, which follows an encoder-decoder paradigm. The encoder consists of sequential convolutional layers with max-pooling operations to reduce the spatial resolution and extract hierarchical features at varying levels of abstraction. Conversely, the decoder reconstructs the feature maps through transposed convolutions (also known as up-sampling layers) to restore the original spatial resolution and generate the final segmentation mask. Skip connections are established between corresponding layers of the encoder and decoder, preserving important spatial information throughout the reconstruction process.

One critical modification we applied to the traditional U-Net is the removal of the final down-sampling layer from the encoder. This adjustment helps to retain more detailed spatial information at the lowest resolution, improving feature extraction without introducing additional parameters. By doing so, our model preserves finer details necessary for precise binary segmentation, which is crucial for medical image analysis where pixel-level precision is required.

B. Dual Attention Mechanisms

To enhance the feature representation capabilities of the U-Net, we incorporate dual attention mechanisms, following the approach introduced by Jun Fu et al. [37]. These mechanisms are composed of two key modules:

1) **Position Attention Module**

This module focuses on the spatial relationships within the feature map, allowing the model to selectively attend to different regions based on their spatial context. This spatial awareness helps in refining the segmentation of object boundaries and enhances the preservation of fine details in the feature maps, ensuring better localization of regions of interest (RoI) in the medical images.

2) **Channel Attention Module**

This module recalibrates the importance of feature channels by dynamically adjusting the contribution of

each channel to the segmentation result. Channels that contain more relevant information for the binary segmentation task are prioritized, improving the model's ability to highlight critical features and suppress irrelevant ones. This selective attention ensures that the most informative features are leveraged during the segmentation process.

These dual attention mechanisms are integrated between the encoder and decoder, serving as a bridge that refines feature maps by leveraging global context information. The combination of spatial and channel attention provides a more robust feature representation, facilitating better segmentation results, especially in complex medical imaging scenarios.

C. Architecture Design and Workflow

The overall architecture of the Dual Attention U-Net is composed of three main components: the encoder, the dual attention module, and the decoder. As illustrated in Figure 1, the model follows the same operational flow as the traditional U-Net, with key improvements.

1) **Encoder**

The encoder is responsible for progressively down-sampling the input image while extracting high-level features. This is done through a series of convolutional layers followed by max-pooling operations. The encoder reduces the spatial resolution and captures hierarchical feature representations from the input images.

2) **Dual Attention Module**

Positioned between the encoder and decoder, the dual attention module processes the output from the encoder. It integrates both the position attention and channel attention mechanisms, allowing the model to focus on the most critical spatial and channel information. The dual attention module applies a global context operation, enriching the feature maps with valuable information that spans beyond local regions. This ensures that the model captures both fine-grained details and broader contextual information before passing the processed feature maps to the decoder.

3) **Decoder**

The decoder performs up-sampling operations to progressively reconstruct the feature maps to their original spatial resolution. It utilizes transposed convolutions to achieve this, while skip connections with the encoder help retain crucial spatial information. The refined feature maps from the dual attention module are passed to the decoder, ensuring that both local and global context are considered during the segmentation process. The final output is a high-resolution binary segmentation mask, where the pixel-level predictions are improved through the informed, context-aware decoding enabled by the dual attention mechanisms.

D. Advantages of Dual Attention U-Net for Binary Segmentation

The integration of dual attention mechanisms enhances the Dual Attention U-Net's ability to capture intricate details and complex patterns within medical images. By focusing on both spatial and channel dimensions, the model improves the quality

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of feature extraction and representation, which is crucial for accurate segmentation. The position attention module ensures that important spatial relationships are preserved, while the channel attention module dynamically emphasizes the most relevant features, improving the overall segmentation accuracy.

Additionally, the removal of the final down-sampling operation in the encoder helps preserve finer details, reducing the loss of critical information that is often essential for medical image analysis. This architecture design effectively balances complexity and performance, ensuring both high precision and efficiency in binary image segmentation tasks.

This Methodology section provides a clear and detailed explanation of the architecture and the novel enhancements introduced with the dual attention mechanisms, aligned with the figure and the description you provided.

IV. EXPERIMENTATION

A. Datasets

The first dataset used in this study is the Eye Disease dataset sourced from Kaggle [44], a renowned platform for sharing and accessing datasets. This dataset originally encompasses a collection of eye disease images, serving as the primary data source for this work. It consists of 383 images, categorized into 5 distinct classes, including Bulging eyes, Cataracts, Crossed eye, Glaucoma, and Uveitis. But each image capture encompasses both eyes: the left and the right. To streamline and focus our analysis, we have opted to crop these images to isolate a single eye. This approach allows us to simplify the visual input by concentrating on individual eyes rather than processing the entire facial context. This cropping technique is particularly relevant in scenarios where the analysis or application requires a more specific examination of the characteristics, movements, or patterns associated with a single eye. By isolating and cropping each image to feature only one eye, we get 1,044 images. And some images without real eye images are excluded from the dataset in the data preprocessing step. Each image has been meticulously labeled by using the LabelMe program, enabling the generation of binary masks that delineate specific regions within the eye images, such as the sclera and pupil areas, as shown in Figure 2.a. This dataset finally consists of 1044 images, where each image is resized to a fixed resolution of 400×400 . In this study, data augmentation is exclusively applied to the eye disease dataset, involving both vertical and horizontal techniques and rotation. This dataset consists of 1044 images, where each image is resized to a fixed resolution of 400×400 . The images are classified into five categories: Bulging eyes, Cataracts, Crossed eye, Glaucoma, and Uveitis. For detailed analysis, each image in the dataset has been carefully labeled with two ground truth annotations: the sclera and pupil area of the eye, using the LabelMe tool. It involves creating binary masks that highlight specific regions within the eye images.

In this study, data augmentation is exclusively applied to the eye disease dataset, involving both vertical and horizontal techniques and rotation. By incorporating these geometric

modifications, the dataset's size has been notably increased, growing from 1044 to 2560 images. This augmentation enhances the richness of the training and validation sets. The same affine transformations applied to the images are also employed on the corresponding masks to maintain data consistency. The deliberate use of these data augmentation techniques provides a sufficient dataset to enhance the model's performance.

The second dataset employed in this study is the Skin Lesion dataset from the International Skin Imaging Collaboration (ISIC) [45]. This dataset is a critical component of deep learning research, focusing on detecting and classifying skin lesions. The Skin Lesion dataset was originally composed of 2,750 images with ground truth labels, as shown in Figure 2.b, encompassing a diverse range of skin lesions and conditions. In providing a valuable resource for evaluating our proposed method, the Skin Lesion dataset after proper modifications includes 2000 images with ground truth labels covering a diverse range of skin lesions and conditions. Each image in the dataset is standardized to a fixed resolution of 256×256 . In our dedication to skin lesion analysis, this dataset has helped develop and evaluate our proposed method, tailored explicitly for binary segmentation tasks. Our method effectively utilizes this dataset to detect and delineate regions of interest within skin lesion images, contributing to enhanced diagnostic accuracy.

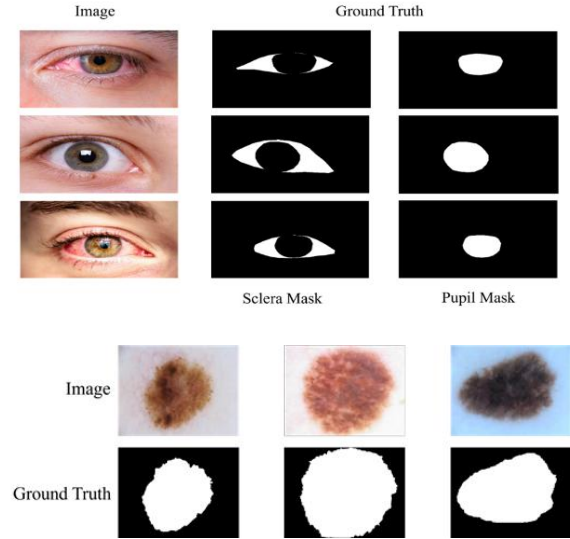


Figure 2. a) The Eye disease dataset with ground truth of sclera mask and pupil mask. b) The Skin lesion dataset with ground truth.

B. Evaluation Metrics

In evaluating the performance of our proposed method, the following evaluation metrics are used: pixel accuracy, dice coefficient, Mean Intersection of Union (mIoU), and sensitivity. These metrics are used to quantify the accuracy between the ground truth and the predicted mask generated by the proposed model, and they are represented by equations 1 through 4, respectively. Pixel accuracy quantifies the ratio of correctly predicted pixels in binary segmentation. mIoU measure overlapping and dice coefficient assess the similarity

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between predicted masks and ground truths. Sensitivity evaluates the model's effectiveness in identifying positive instances in binary segmentation. The mathematical equations for each metric are as follows:

$$\text{Pixel Accuracy} = \frac{TP + TN}{TP + FN + FP + TN} \quad (1)$$

$$\text{mIoU} = \frac{TP}{TP + FP + FN} \quad (2)$$

$$\text{Dice Coefficient} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (3)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (4)$$

Where TP represents true positives, TN signifies true negatives, FP corresponds to false positives, and FN indicates false negatives.

C. Implementation

The proposed model experienced training using 80 percent of the data for training purposes, with the remaining 20 percent reserved for testing. Evaluation metrics were computed for the estimated mask and the ground truth across both datasets. Various hyperparameters have been carefully chosen to train our proposed models, employing a structured approach. The training process defined 50 epochs with a learning rate of $1e-5$, a weight decay of 0.001, and a batch size of 8, utilizing the Adam optimizer and Focal loss function. Additionally, a dropout rate of 0.2 is applied in the encoder block to prevent overfitting. For normalizing the output, a Sigmoid classifier is employed. Image sizes are scaled to the unique characteristics of the data, with dimensions set at 400×400 for eye disease images and 256×256 for skin lesion images. Data augmentation techniques, such as rotation (30 degrees) and both vertical and horizontal flips, are incorporated to enhance model robustness. The PyTorch deep learning framework serves as the backbone for implementing our proposed model. The refined configuration of fine-tuning parameter setting is comprehensively outlined in Table 1.

TABLE I
THE SUMMARY OF THE FINE-TUNING PARAMETER SETTING.

Setting	Value
DL framework	PyTorch
Optimizer	Adam
Activation function	Sigmoid
Learning rate	$1e-5$
Weight decay	0.001
Dropout	0.2
Thresholding	0.5
Batch size	8
Epoch	50
Image size (Eye image)	[400×400]
Image size (Skin lesion)	[256×256]
Data augmentation	Rotation (30 degrees), horizontal flip, vertical flip

IV. RESULTS AND EVALUATION

A. Performance Result of Proposed Method

The evaluation of our proposed model stands as a critical section in this study, where we investigate our model's performance across two distinct testing datasets: eye disease and skin lesion. This evaluation aims to explore the model's efficacy in accurately classifying and segmenting appropriate features within these testing datasets. Our approach to this assessment involves the implementation of carefully selected performance metrics tailored to the specific characteristics of our proposed model. Notably, the eye disease dataset poses unique challenges, demanding precise estimation of two distinct areas, the sclera and pupil area, through estimated segmentation. On the other hand, the skin lesion dataset requires intricate delineation of Regions of Interest (ROI) within images. By exploring these intricacies, our evaluation seeks to provide a comprehensive understanding of our model's capabilities in handling diverse segmentation tasks across different medical imaging domains.

TABLE IV
THE EVALUATION RESULT OF THE PROPOSED MODEL ON BOTH TESTING DATASETS

Datasets	Loss	PA	mIoU	Dice	Sensitivity
Eye Disease	0.002	98.19	92.67	93.00	93.31
Skin Lesion	0.002	97.21	96.55	89.22	90.06

As shown in Table 4, The evaluation of our model on the eye disease dataset reveals promising results. With a learning loss of 0.002, the model demonstrates effective convergence during training. Notably, the high pixel accuracy of 98.19% indicates precise pixel-level classification, showcasing the model's proficiency in distinguishing features within the eye images. The mIoU of 92.67% further underscores the model's strong segmentation performance, ensuring accurate delineation of segmented areas. In addition, the high Dice coefficient of 93.00% reflects substantial agreement between predicted and ground truth segmentations, affirming the model's accuracy. The sensitivity of 93.31% highlights the model's effectiveness in capturing positive instances within the eye disease dataset.

In the evaluation of the skin lesion dataset, the model continues to exhibit commendable performance, as illustrated in Table 4. The learning loss of 0.002 signifies effective training convergence, while the high pixel accuracy of 97.21% demonstrates accurate pixel-level classification. The mIoU of 96.55% emphasizes the model's proficiency in precise delineation of segmented regions within skin lesion images. Although the Dice coefficient of 89.22% is slightly lower than in the eye disease dataset, it still indicates substantial agreement between predicted and ground truth segmentations. Moreover, the sensitivity of 90.06% underscores the model's capability to detect positive instances within the skin lesion dataset. Overall, these results highlight the versatility and effectiveness of our proposed model in handling diverse segmentation tasks across distinct medical imaging domains.

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Our evaluation process involved a visual examination of selected sample images from both the eye disease and skin lesion datasets. In this analysis, we compared the binary masks predicted by our proposed model along with the existing methods to the ground truth, as illustrated in including Figure 7 and Figure 8 for the eye disease datasets and Figure 9 and Figure 10 for the skin lesion dataset. These visualizations offer a comprehensive understanding of the model's performance, highlighting its effectiveness compared to existing methods.

Through this visual investigation, specific instances where our model achieved exceptional accuracy and areas where it encountered challenges or exhibited inaccuracies were identified. Notably, the outcome visualizations revealed suboptimal performance of existing models on the dataset, except for U-Net, which demonstrated satisfactory results on eye disease images, and Deeplabv3plus, which indicated commendable estimated results on skin lesion images. However, our proposed method surpassed the performance of both U-Net and Deeplabv3plus, showcasing efficacy and emphasizing its potential for achieving heightened accuracy in segmenting both image datasets.

B. Performance Comparison with Previous Methods

In the result, we show a score of the performance of our proposed method and a comparative analysis of the alongside established baseline models, including U-Net [31], DeepLabv3 [46], DeepLabv3plus [47], Residual U-Net [48], and dual attention network [37]. Our evaluation metrics were conducted on two critical medical datasets: the eye disease and skin lesion datasets. Focal loss, pixel accuracy, mIoU, dice coefficient, and sensitivity evaluation methods were utilized to analyze the model's performance, shown as a graph of comparison between existing models and our proposed model on both datasets.

As shown in Table 2, our proposed method stands out as the top performer on the eye disease dataset. It achieved a minimal loss of 0.001, which indicates that our proposed model learned effectively on the eye disease regions. Pixel accuracy of 99.01% suggests that the proposed model correctly identifies almost all the pixels, which demonstrates exceptional accuracy in distinguishing eye disease areas. The mIoU score of 96.13% shows a substantial overlap between the model's predicted mask and ground truth, which indicates precision in accurate overlapping. A dice coefficient score of 96.57% indicates the model's effectiveness in capturing eye disease regions while minimizing false positives. The last one has a sensitivity of 96.76%, which suggests that the model effectively recognizes nearly all the actual regions in the images. In the comparison of the eye disease dataset, existing models, including U-Net, Res-Net, Attention-Unet, DeepLabv3, DeepLabv3plus, and DANet, demonstrated commendable performance. Exceptionally, U-Net and DeepLab3plus performed well in terms of loss and pixel accuracy. However, none of them achieved the mIoU, dice coefficient, and sensitivity levels attained by our proposed method. These results underscore the superiority of our method in achieving precise and sensitive binary segmentation of eye disease images.

TABLE II
RESULT OF THE PROPOSED MODEL AND EXISTING MODEL ON THE EYE DISEASE DATASET

Model	Loss	PA	mIoU	Dice	Sensitivity
U-Net [31]	0.001	95.87	87.52	85.90	84.40
Res-Net [48]	0.009	92.02	90.64	78.01	76.44
Attention Unet	0.010	92.66	79.48	79.49	79.30
DeepLabv3 [46]	0.006	92.16	93.42	81.90	80.38
DeepLabv3+ [47]	0.002	97.24	97.94	94.40	94.41
DANet [37]	0.140	90.24	88.74	72.52	70.00
Proposed Net	0.001	98.57	98.20	96.17	96.89

TABLE III
RESULT OF THE PROPOSED MODEL AND EXISTING MODEL ON THE SKIN LESION DATASET

Model	Loss	PA	mIoU	Dice	Sensitivity
U-Net [31]	0.001	96.63	87.64	88.06	86.85
Res-Net [48]	0.007	92.30	73.72	78.08	65.41
Attention Unet	0.036	91.13	71.02	69.16	70.46
DeepLabv3 [46]	0.009	93.79	78.75	77.32	75.85
DeepLabv3+ [47]	0.015	94.01	79.30	78.27	76.75
DANet [38]	0.150	89.93	71.05	67.03	69.43
Proposed Net	0.001	99.01	96.13	96.57	96.76

As illustrated in Table 3, we present the result of the proposed model compared to existing approaches on skin lesion datasets. The proposed model is also well-performed on this dataset. Our model outperformed all other models with a minimal loss of 0.001, which indicates that our proposed method's segmentation is very close to the ground truth, signifying the model's high precision and accuracy in identifying skin lesion regions. Achieving a pixel accuracy of 98.57% indicates that our method correctly estimated almost all the pixels, which signifies a high degree of accuracy in distinguishing between skin lesions and non-lesion areas. A mIoU score of 98.20% reflects a substantial overlap between the predicted mask and ground truth. For dice coefficients, the score is 96.17%, which means that our model effectively captures nearly all the actual skin lesion regions in the images. In the comparison, the existing models, such as U-Net, Res-Net, Attention-Unet, DeepLabv3, DeepLabv3+, and DANet, expressed commendable performance but fell short of the proposed method's results. In particular, the U-Net model achieved a pixel accuracy of 95.87% and a mIoU score of 87.52%, but our method outperformed it in both metrics. Similar trends were observed with other existing models, with the proposed approach consistently surpassing them in pixel accuracy, mIoU, dice coefficient, and sensitivity. These results underscore the robustness of the proposed method in binary segmentation tasks on the skin lesion dataset.

VI. DISCUSSION

The results obtained from evaluating our proposed method on both eye disease and skin lesion training and testing datasets are highly encouraging. The comprehensive performance

metrics, encompassing low loss values, better pixel accuracy, mIoU, dice coefficients, and sensitivity scores, collectively underscore the model's accomplishment in binary segmentation tasks for medical image analysis. These findings strongly suggest that our proposed method excels in the intricate tasks of accurate and precise pixel-level segmentation and classification.

To further validate the efficacy of our proposed method, we conducted a comparative analysis against established methods within the realm of medical image segmentation. This comparative assessment consistently revealed that our model outperforms several existing methods across crucial metrics such as mIoU, pixel accuracy, dice coefficient, and sensitivity. Noteworthy is our model's significant improvement in sensitivity scores, a significant aspect of medical image analysis for ensuring the correct identification of ROI. These results compellingly emphasize the superiority of our approach in the specific context of eye disease and skin lesion segmentation.

While our proposed method demonstrates remarkable performance in binary segmentation tasks, it is crucial to acknowledge certain limitations. The efficacy of the method heavily depends on the availability of diverse and representative training data. Its ability to generalize to real-world images may be constrained if the dataset fails to capture a wide range of scenarios and variations in application conditions. Additionally, the computational demands associated with deep learning can be substantial, potentially limiting deployment in resource-constrained settings or applications requiring real-time processing.

The interpretability of our model is another consideration, given that complex deep learning architectures often lack transparency. It can pose challenges in scenarios where understanding the model's segmentation decisions is essential. The risk of overfitting is also a concern, mainly as the method achieves high accuracy on training data. Strategies to mitigate overfitting, such as data augmentation and regularization techniques, may be necessary. Lastly, ensuring the method's ability to deliver fast and accurate segmentation results in real-time applications, especially in the presence of complex image pixels, is crucial. The extent to which the method generalizes to new and unseen domains beyond the training dataset remains an ongoing area of research.

VI. CONCLUSION

In conclusion, our thesis represents a significant contribution to medical image analysis. We introduced a highly effective deep learning model tailored for binary segmentation tasks, explicitly focusing on eye disease and skin lesion datasets. Our model's robust accuracy and precision underscore its suitability for pixel-level classification and segmentation. Our model consistently demonstrated superiority through a rigorous comparative analysis with existing methods, setting a benchmark in the domain. The clinical relevance of our research is highlighted by the high sensitivity scores, positioning it as a valuable tool for accurate region of interest identification, thereby supporting the diagnosis and treatment

of eye diseases and skin lesions.

The reach of our research extends far beyond the current study. By showcasing the potential of deep learning in medical image analysis, we offer a robust framework for precise segmentation, with the capacity to revolutionize clinical practice and provide more accurate diagnoses. The comparative analysis underscores the advancements our research brings to the field, promoting higher image segmentation accuracy and clinical relevance standards. Looking toward the future, we see promising opportunities for extending our research. We plan to broaden the scope of our model to overcome semantic and instance segmentation tasks, offering more comprehensive and delicate medical image analysis and automotive detection. This extension will involve accommodating multiple classes and distinguishing between individual instances, objects, and backgrounds within an image, ultimately facilitating advanced and detailed analysis in clinical fields and other industry domains.

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